SWE 370 - Data Mining Project

Project Purpose & Learning Outcomes:

The aim of this project is to help you understand and apply the concepts of classification, impurity measurement, and decision tree splitting using real datasets.

REMEMBER THIS PROJECT IS WORTH 40% OF YOUR GRADE!!

Project Requirements:

1. Dataset Selection

- Choose any publicly available dataset
- Minimum: **100 records**
- Minimum: **3 attributes (features)**
- Clearly identify the **target class label**

Recommended sources:

- Kaggle
- UCI Machine Learning Repository

2. Impurity & Gain Calculations

For **two chosen attributes**:

- Compute impurity before splitting (P)
- Compute weighted impurity after splitting (M)
- Compute gain: Gain = P – M
- Decide which attribute is best

3. Python Implementation

- Load dataset
- Compute P, M, and gain
- Understand the code
- Include **screenshots** of key code parts in the report

4. Report Content

- Dataset description
- Calculations for P, M, and gain
- Python screenshots and brief explanation
- Final conclusion (which attribute is best and why)

5. Presentation Requirements

- Each group must prepare a **20–30 minutes presentation**
- All calculations must be shown clearly
- Python results must be discussed
- Every student must be ready to answer questions

Team Structure:

- Each group must consist of **7 students**
- All members must actively participate in:
  - ✓ The analysis of the dataset
  - ✓ The Python implementation
  - ✓ Preparing the written report
  - ✓ Preparing the presentation
- Each group will have **one team leader** responsible for weekly coordination

Each group leader must upload the project **ZIP file** on UZEM before the deadline. **No extensions will be made.**

Project file must be named in the following format: GroupNumber_ProjectName

Project file must contain:

1. PDF report (8–12 pages)
2. Python code file (.py or notebook)
3. PowerPoint presentation

Follow up announcements to know your presentation date.